

Kubernetes Deployment:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ minikube start
* minikube v1.38.1 on Microsoft windows 11 Pro 25H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Updating the running docker "minikube" container ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ vim deployment.yaml
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ ls
deployment.yaml
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-deploy
spec:
  replicas: 2
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
        - name: nginx
          image: nginx:1.25
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl apply -f deployment.yaml
deployment.apps/web-deploy created
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
web-deploy	2/2	2	2	6m57s

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get rs
```

NAME	DESIRED	CURRENT	READY	AGE
web-deploy-55bb7667c5	0	0	0	7m9s
web-deploy-86648cb98c	2	2	2	23s

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
web-deploy-86648cb98c-7f88p	1/1	Running	0	50s
web-deploy-86648cb98c-jwwxs	1/1	Running	0	49s

Rolling updates:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ vim deployment.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ ls
deployment.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-deploy
spec:
  replicas: 2
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
        - name: nginx
          image: nginx:1.26
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl apply -f deployment.yaml
deployment.apps/web-deploy configured

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
web-deploy    2/2     2            2           15m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get rs
NAME                                DESIRED   CURRENT   READY   AGE
web-deploy-55bb7667c5              0         0         0       15m
web-deploy-59884c654b              2         2         2       25s
web-deploy-86648cb98c              0         0         0       8m22s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get pods
NAME                                READY     STATUS    RESTARTS   AGE
web-deploy-59884c654b-6xhm4        1/1       Running   0          57s
web-deploy-59884c654b-lsbtw        1/1       Running   0          56s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl rollout status deployment web-deploy
deployment "web-deploy" successfully rolled out
```

Rollback Bad Deployment:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl rollout history deployment web-deploy
deployment.apps/web-deploy
REVISION   CHANGE-CAUSE
1          <none>
2          <none>
3          <none>

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl rollout history deployment web-deploy --revision=2
deployment.apps/web-deploy with revision #2
Pod Template:
  Labels:  app=web
           pod-template-hash=86648cb98c
  Containers:
    nginx:
      Image:  nginx:1.25
      Port:  <none>
      Host Port: <none>
      Environment: <none>
      Mounts: <none>
      Volumes: <none>
      Node-Selectors: <none>
      Tolerations: <none>

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl rollout undo deployment web-deploy
warning: resource deployments/web-deploy was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration
annotation, which may cause unexpected behavior on future 'kubectl apply' operations. consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/web-deploy rolled back
```

Verify:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3
$ kubectl get rs
NAME                                DESIRED   CURRENT   READY   AGE
web-deploy-55bb7667c5              0         0         0       23m
web-deploy-59884c654b              0         0         0       9m12s
web-deploy-86648cb98c              2         2         2       17m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3
$ kubectl rollout status deployment web-deploy
deployment "web-deploy" successfully rolled out
```

Scaling:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ vim deployment1.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ ls
deployment.yaml  deployment1.yaml
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl apply -f deployment1.yaml
deployment.apps/web-deploy configured

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ ^C

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
web-deploy    3/3     3            3           77m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl scale deployment web-deploy --replicas=4
deployment.apps/web-deploy scaled

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
web-deploy    4/4     4            4           84m
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl describe deployment web-deploy
Name:          web-deploy
Namespace:     default
CreationTime:  Wed, 05 Aug 2026 11:42:35 +0530
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 4
Selector:      app=web
Replicas:      4 desired | 4 updated | 4 total | 4 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=web
  Containers:
    nginx:
      Image:      nginx:1.25
      Port:       <none>
      Host Port:  <none>
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations:  <none>
  Conditions:
    Type           Status  Reason
    ----           -
    Available       True    MinimumReplicasAvailable
    Progressing     True    NewReplicaSetAvailable
OldReplicaSets:  web-deploy-55bb7667c5 (0/0 replicas created), web-deploy-59884c654b (0/0 replicas created)
NewReplicaSet:   web-deploy-86648cb98c (4/4 replicas created)
Events:
  Type     Reason          Age    From          Message
  ----     -
  Normal   ScalingReplicaSet   64m (x2 over 77m)    deployment-controller   Scaled up replica set web-deploy-86648cb98c from 0 to 1
  Normal   ScalingReplicaSet   64m (x2 over 77m)    deployment-controller   Scaled up replica set web-deploy-86648cb98c from 1 to 2
  Normal   ScalingReplicaSet   64m (x2 over 64m)    deployment-controller   (combined from similar events): Scaled down replica set web-deploy-59884c654b from 1 to 0
  Normal   ScalingReplicaSet   9m39s    deployment-controller   Scaled up replica set web-deploy-86648cb98c from 2 to 3
  Normal   ScalingReplicaSet   47s      deployment-controller   Scaled up replica set web-deploy-86648cb98c from 3 to 4
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl rollout pause deployment web-deploy
deployment.apps/web-deploy paused

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
web-deploy    4/4     4            4           89m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl rollout resume deployment web-deploy
deployment.apps/web-deploy resumed

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get rs
NAME                                DESIRED   CURRENT   READY   AGE
web-deploy-55bb7667c5              0         0         0       89m
web-deploy-59884c654b              0         0         0       74m
web-deploy-86648cb98c              4         4         4       82m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-deploy-86648cb98c-5k7x4        1/1     Running   0          69m
web-deploy-86648cb98c-6cxh4        1/1     Running   0          69m
web-deploy-86648cb98c-nf9s8        1/1     Running   0          6m26s
web-deploy-86648cb98c-wm2zh        1/1     Running   0          15m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl delete deployment web-deploy
deployment.apps "web-deploy" deleted from default namespace

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-3 (main)
$ kubectl get pods
No resources found in default namespace.
```